Global Hypoellipticity of a Vector Field Implies Zero Entropy

Abstract

We prove that a smooth vector field X on a compact connected smooth manifold whose Lie derivative is globally hypoelliptic on distributions generates a flow of zero topological entropy. The proof uses two standard results: the Greenfield–Wallach–Chen–Chi–Forni theorem that globally hypoelliptic vector fields are cohomology-free, and Katok's entropy obstruction to finite-dimensional stable smooth cohomology.

1 Preliminaries

Let M be a compact connected smooth manifold and let $X \in \mathfrak{X}^\infty(M)$. Write g_t for the flow of X, which is complete because M is compact. Let $\mathcal{D}'(M)$ denote the space of distributions on M. We use the convention that $Xf = \mathcal{L}_X f$ for $f \in C^\infty(M)$, and the same symbol denotes the transpose action on distributions.

Definition 1. *The vector field X is called* globally hypoelliptic *if*

$$u \in \mathcal{D}'(M), \quad Xu \in C^\infty(M) \quad \Longrightarrow \quad u \in C^\infty(M).$$

*It is called C^∞-*cohomology-free *if for every $f \in C^\infty(M)$ there exist $u \in C^\infty(M)$ and a constant $c \in \mathbb{C}$ such that*

$$Xu = f - c.$$

We shall use the following two standard theorems.

Theorem 1 (Greenfield–Wallach; Chen–Chi; Forni)**.** *Let M be a compact connected smooth manifold and let $X \in \mathfrak{X}^\infty(M)$. If X is globally hypoelliptic, then X is C^∞-cohomology-free. Moreover, the flow of X preserves a smooth probability density μ, the space of X-invariant distributions is one-dimensional, namely*

$$\ker(X : \mathcal{D}'(M) \to \mathcal{D}'(M)) = \mathbb{C}\mu,$$

and g_t is uniquely ergodic.

Reference. This is the equivalence between global hypoellipticity and cohomology-freeness proved in Greenfield–Wallach [3], refined by Chen–Chi [1], and stated in the above form by Forni [2, Theorem 3.10]. Forni's formulation does not require orientability if smooth densities are used instead of volume forms. □

The second input is Katok's entropy obstruction to finite-dimensional stable smooth cohomology.

Theorem 2 (Katok's entropy obstruction)**.** *Let $X \in \mathfrak{X}^\infty(M)$ generate a smooth flow g_t on a compact manifold. Suppose the range*

$$X(C^\infty(M)) \subset C^\infty(M)$$

is closed and has finite codimension. Then

$$h_{\mathrm{top}}(g_t) = 0.$$

Equivalently, if $h_{\mathrm{top}}(g_t) > 0$, then the smooth cohomology

$$C^\infty(M)/X(C^\infty(M))$$

is infinite-dimensional or the range $X(C^\infty(M))$ is not closed.

Reference. This is a standard consequence of Katok's combinatorial construction of smooth cocycles over positive-entropy systems. In the form used here, it appears in Katok–Robinson [4, Section 6] and Katok [5, Chapter 6]: positive topological entropy produces arbitrarily many linearly independent smooth cocycles modulo smooth coboundaries. Thus a C^∞-stable flow with finite-dimensional smooth cohomology must have zero topological entropy. □

2 Proof of the theorem

Theorem 3. *Let M be a compact connected smooth manifold and let $X \in \mathfrak{X}^\infty(M)$. Assume that whenever $f \in C^\infty(M)$, $u \in \mathcal{D}'(M)$, and*

$$Xu = f$$

in the sense of distributions, then $u \in C^\infty(M)$. Then the flow g_t generated by X has zero topological entropy.

Proof. The stated hypothesis is precisely global hypoellipticity of the Lie derivative $X = \mathcal{L}_X$ on distributions. By Theorem 1, X is C^∞-cohomology-free. Hence for every $f \in C^\infty(M)$ there are $u \in C^\infty(M)$ and $c \in \mathbb{C}$ such that

$$f = Xu + c.$$

Therefore

$$C^\infty(M) = X(C^\infty(M)) + \mathbb{C} \cdot 1.$$

Let μ be the smooth invariant probability density supplied by Theorem 1. Since $X\mu = 0$, we have

$$\int_M Xu \, d\mu = 0 \qquad \text{for every } u \in C^\infty(M).$$

Consequently,

$$X(C^\infty(M)) \subset \left\{ f \in C^\infty(M) : \int_M f \, d\mu = 0 \right\}.$$

The cohomology-free property gives the reverse inclusion: if $\int_M f \, d\mu = 0$, then in the decomposition $f = Xu + c$ we must have $c = 0$. Thus

$$X(C^\infty(M)) = \left\{ f \in C^\infty(M) : \int_M f \, d\mu = 0 \right\}.$$

The right-hand side is a closed hyperplane in the Fréchet space $C^\infty(M)$. Hence $X(C^\infty(M))$ is closed and has codimension one.

Katok's entropy obstruction, Theorem 2, now applies and yields

$$h_{\mathrm{top}}(g_t) = 0.$$

Equivalently, $h_{\mathrm{top}}(g_1) = 0$, and by the usual scaling law for flows, $h_{\mathrm{top}}(g_t) = |t|\, h_{\mathrm{top}}(g_1) = 0$ for every $t \in \mathbb{R}$. □

Remark 1. *The proof uses substantially more than unique ergodicity. Unique ergodicity alone does not force zero topological entropy in general topological dynamics. The essential point is the finite-dimensional, closed C^∞-cohomology forced by global hypoellipticity.*

References

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